

Brac University
Department of Electrical & Electronic Engineering
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Course Number : EEE282

Section : 01

Group No : 04

Lab Report

Experiment no: 09

Name of the experiment:

Linear Ordinary Differential Equations

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Objective:

The objective of this experiment is to solve first-order differential equations using Euler's method and Improved Euler's method, and to compare the numerical results with the exact analytical solutions. This helps us understand the accuracy of numerical methods and the effect of step size on the solution.

Theoretical Background:

Euler's method is one of the simplest numerical techniques for solving first-order differential equations. It works by using the slope of the solution at a known point to estimate the value at the next point. Starting from an initial condition, the solution is advanced step-by-step using a fixed step size h . Although Euler's method is easy to implement, it can produce significant errors, especially when the step size is large, because it only considers the slope at the beginning of each interval.

To reduce this error, the Improved Euler method is used. This method first predicts the next value using Euler's method and then corrects it by taking the average of the slopes at the current point and the predicted point. By considering more information about the slope, the Improved Euler method provides better accuracy than the simple Euler method.

Problem Statement:

Exercise-1:

Exercise 1. Use Euler's method to solve the IVP $y' = \frac{x-y}{2}$ on $[0, 3]$ with $y(0) = 1$.

Compare solutions for $h = 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}$

The exact solution is $y = 3e^{-x/2} - 2 + x$

Answer:

Code:

```
clear all;
close all;
clc;

for h = [0.125, 0.25, 0.5, 1]

    x = 0:h:3;
    y = zeros(1,length(x));
    y(1)= 1;

    for n=2:length(x)

        y(n) = y(n-1)+ h.*((x(n-1)-y(n-1))./2);
    end
    fprintf('for h = %f\n',h)
    disp('          x      Euler y')
    disp([x'      y'])
    fprintf('\n')

y_exact = 3*exp(-x/2) - 2 + x;
```

```

% Plot
figure
    plot(x,y)
    hold on
    plot(x,y_exact)
    grid on
    legend('Euler','Exact Solution')
    title(['h = ',num2str(h)])
end

```

Output:

```

for h = 0.125000
    x    Euler y
    0    1.0000
    0.1250  0.9375
    0.2500  0.8867
    0.3750  0.8469
    0.5000  0.8174
    0.6250  0.7976
    0.7500  0.7868
    0.8750  0.7845
    1.0000  0.7902
    1.1250  0.8033
    1.2500  0.8234
    1.3750  0.8500
    1.5000  0.8829
    1.6250  0.9214
    1.7500  0.9654
    1.8750  1.0144
    2.0000  1.0682
    2.1250  1.1265
    2.2500  1.1889
    2.3750  1.2552
    2.5000  1.3252
    2.6250  1.3986
    2.7500  1.4753
    2.8750  1.5549
    3.0000  1.6374
end

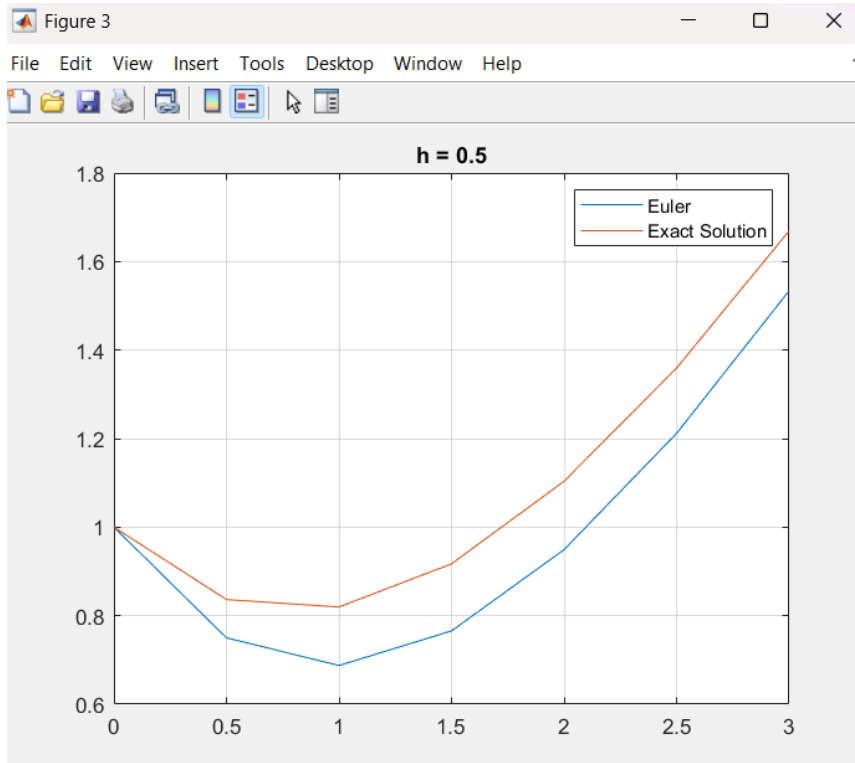
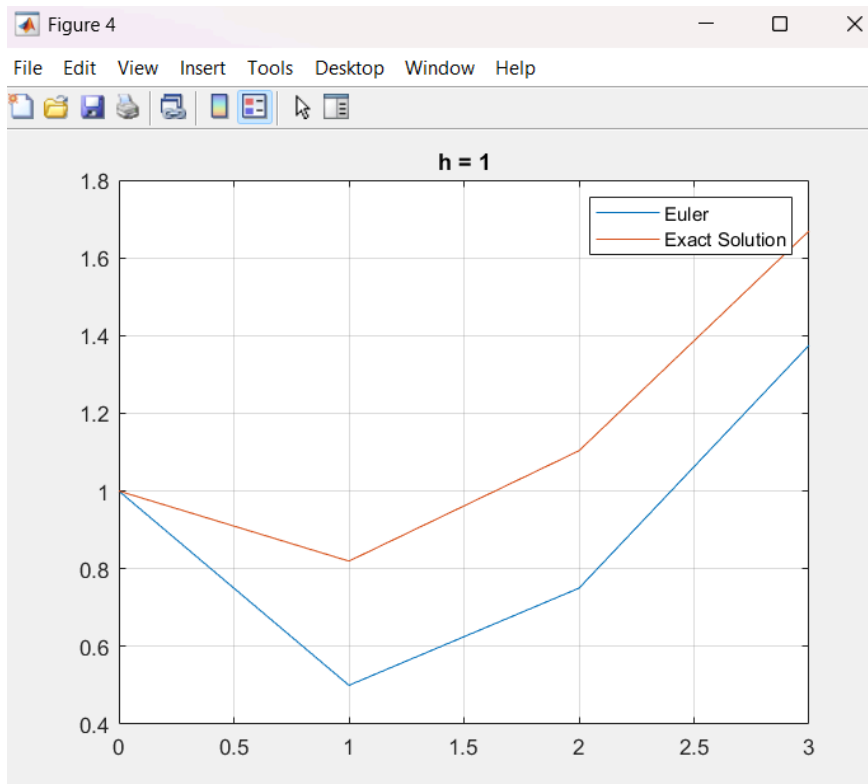
for h = 0.250000
    x    Euler y
    0    1.0000
    0.2500  0.8750
    0.5000  0.7969
    0.7500  0.7598
    1.0000  0.7585
    1.2500  0.7887
    1.5000  0.8464
    1.7500  0.9281
    2.0000  1.0308
    2.2500  1.1520
    2.5000  1.2892
    2.7500  1.4406
    3.0000  1.6043
end

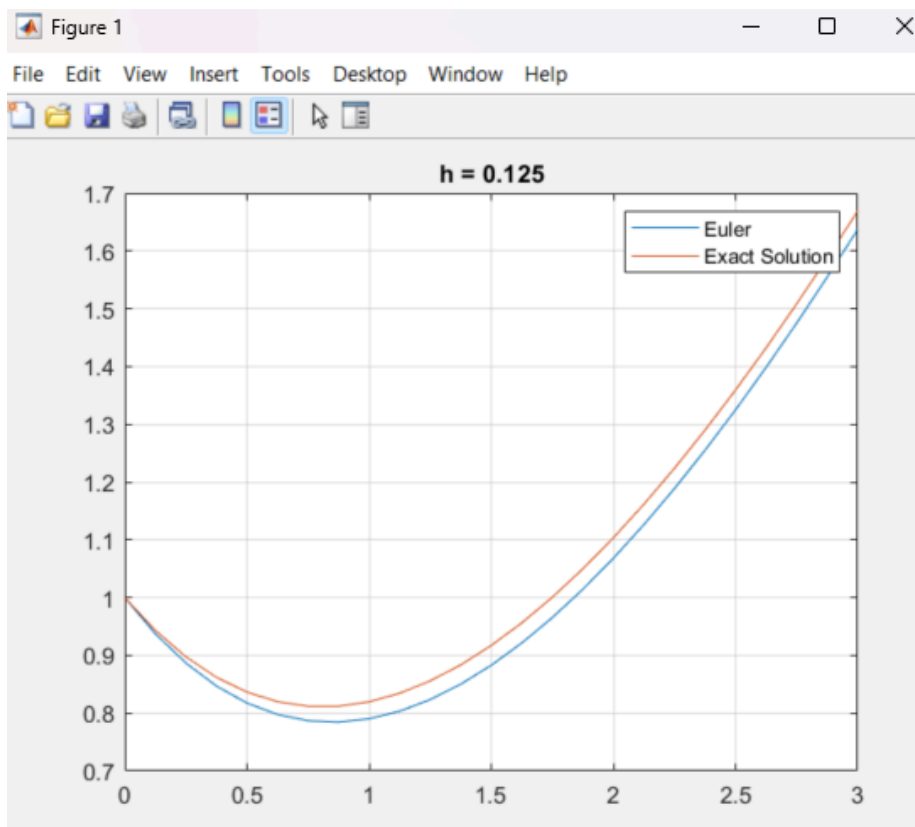
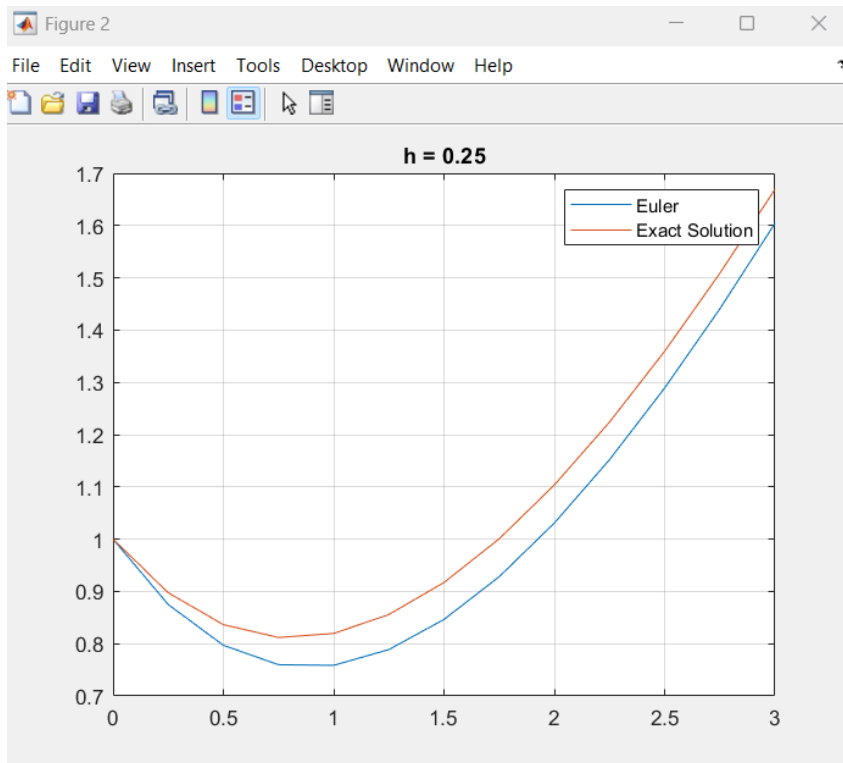
for h = 1.000000
    x    Euler y
    0    1.0000
    1.0000  0.5000
    2.0000  0.7500
    3.0000  1.3750
end

for h = 0.500000
    x    Euler y
    0    1.0000
    0.5000  0.7500
    1.0000  0.6875
    1.5000  0.7656
    2.0000  0.9492
    2.5000  1.2119
    3.0000  1.5339
end

```

Graphs: (Graph between Euler and Exact Solution)



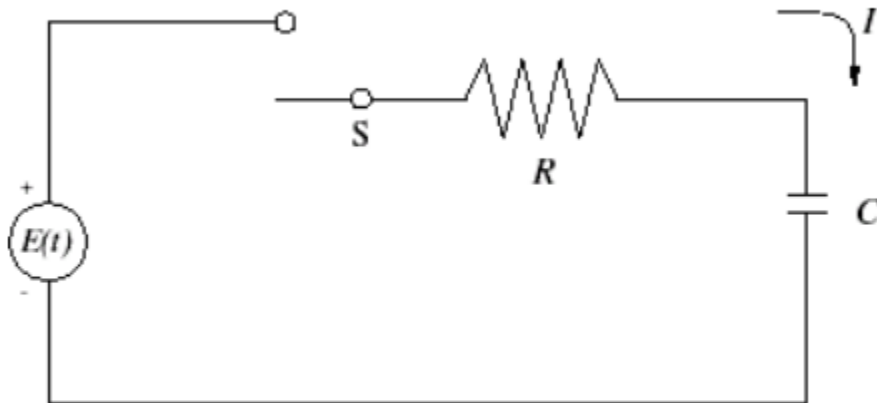


Explanation:

This code solves a first-order differential equation using Euler's method and compares the result with the exact (analytical) solution. The program repeats the calculation for four different step sizes ($h = 0.125, 0.25, 0.5, 1$) to show how step size affects accuracy. For each value of h , the code creates the x values from 0 to 3, sets the initial condition $y(1) = 1$, and then computes the solution step-by-step using Euler's formula based on the slope $(x-y)/2$. After computing the numerical solution, it prints the results, calculates the exact solution $y=3e^{-x/2}-2+x$ and plots both solutions on the same graph. The plots clearly show that smaller step sizes give a numerical solution that is closer to the exact solution, while larger step sizes produce more error.

Exercise-2:

Exercise 2. Consider the following circuit:



In this circuit, $R = 20K\Omega$, $C = 10\mu F$, $E = 117 V$ and $Q(0) = 0$. Find Q for $t = 0$ to $t = 3sec$.

Answer:

Code:

```
clear all;
close all;
clc;

R = 20e3;
C = 10e-6;
E = 117;

h = 0.1;           % step size
t = 0:h:3;
Q = zeros(1, length(t));
Q(1) = 0;

% Euler's Method
for n = 2:length(t)
```

```

    Q(n) = Q(n-1) + h * (E/R - Q(n-1) / (R*C));
end

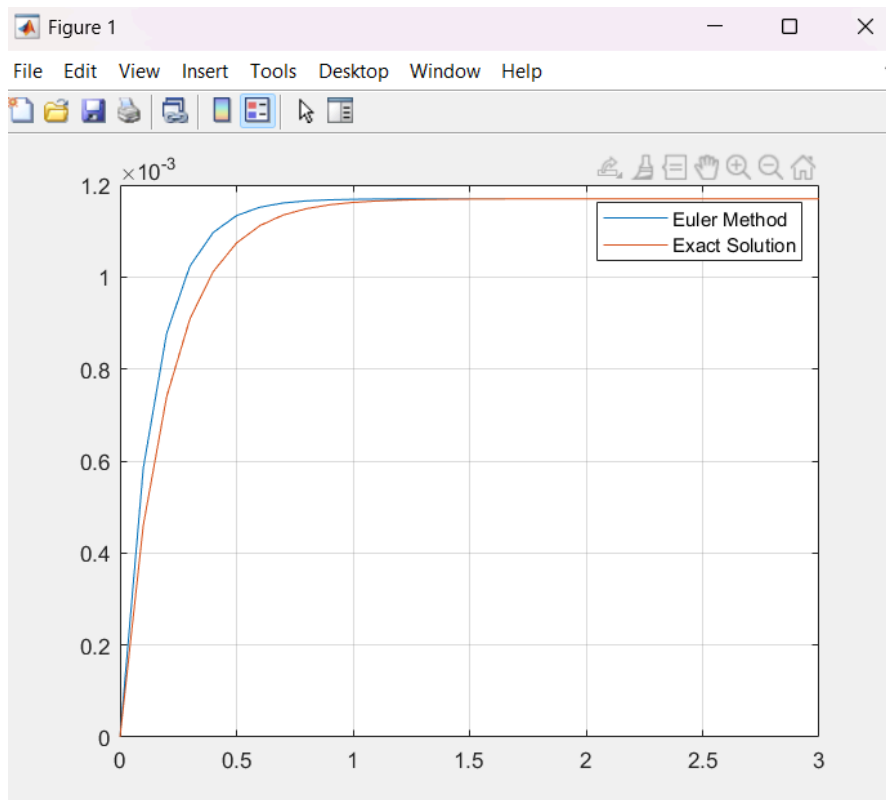
disp('    t        Euler Q');
disp([t' Q']);

% Exact solution
Q_exact = C*E*(1 - exp(-t/(R*C)));

plot(t, Q)
hold on
plot(t, Q_exact)
grid on
legend('Euler Method', 'Exact Solution')

```

Output:



Command Window		
t	Euler Q	
0	0	
0.1000	0.0006	
0.2000	0.0009	
0.3000	0.0010	
0.4000	0.0011	
0.5000	0.0011	
0.6000	0.0012	
0.7000	0.0012	
0.8000	0.0012	
0.9000	0.0012	
1.0000	0.0012	
1.1000	0.0012	
1.2000	0.0012	
1.3000	0.0012	
1.4000	0.0012	
1.5000	0.0012	
1.6000	0.0012	
1.7000	0.0012	
1.8000	0.0012	
1.9000	0.0012	
2.0000	0.0012	
2.1000	0.0012	
2.2000	0.0012	
2.3000	0.0012	
2.4000	0.0012	
2.5000	0.0012	
2.6000	0.0012	
2.7000	0.0012	
2.8000	0.0012	
2.9000	0.0012	
3.0000	0.0012	

Explanation:

In this, code uses Euler's method to find the charge $Q(t)$ in an RC circuit and compares it with the exact solution. First, the resistance R , capacitance C , and voltage E are defined, and a step size $h=0.1$ is chosen. The time values are created from 0 to 3 seconds, and the initial charge is set to zero. Using Euler's formula, the code calculates the charge at each time step based on the circuit equation, where the change in charge depends on the supply voltage and the current charge on the capacitor. The results are printed in a table showing time and Euler's solution. Then, the exact solution $Q=CE(1-e^{-t/(RC)})$ is computed and plotted together with the Euler solution. The graph shows how closely Euler's method follows the exact curve, and it helps us understand how the capacitor charges over time in an RC circuit.

Exercise-3:

Solve exercise 1 and 2 using Improved Euler's method.

Answer:

Exercise 1(Improved Euler's method)

Code:

```
clear all;
```

```
close all;
```

```
clc;
```

```
for h = [0.125, 0.25, 0.5, 1]
```

```
    x = 0:h:3;
```

```
    y = zeros(1,length(x));
```

```
    y(1)= 1;
```

```
    for n=2:length(x)
```

```
        y_pred = y(n-1)+ h.*((x(n-1)-y(n-1))./2);
```

```
        y(n)=y(n-1) +(h/2)*((x(n-1)-y(n-1)/2)+((x(n-1)-y_pred)/2));
```

```
    end
```

```
    fprintf('for h = %f\n',h)
```

```
    disp('      x   Euler y')
```

```
    disp([x'   y'])
```

```
fprintf('\n')
```

```
y_exact = 3*exp(-x/2) - 2 + x;
```

```
% Plot
```

```
figure
```

```
plot(x,y)
```

```
hold on
```

```
plot(x,y_exact)
```

```
grid on
```

```
legend('Euler','Exact Solution')
```

```
title(['h = ',num2str(h)])
```

```
end
```

Output:

for h = 0.125000

x	Euler y
0	1.0000
0.1250	0.9395
0.2500	0.8940
0.3750	0.8629
0.5000	0.8450
0.6250	0.8398
0.7500	0.8463
0.8750	0.8639
1.0000	0.8919
1.1250	0.9297
1.2500	0.9767
1.3750	1.0323
1.5000	1.0960
1.6250	1.1674
1.7500	1.2459
1.8750	1.3311
2.0000	1.4226
2.1250	1.5201
2.2500	1.6231
2.3750	1.7314
2.5000	1.8445
2.6250	1.9624
2.7500	2.0845
2.8750	2.2107
3.0000	2.3408

for h = 0.250000

x	Euler y
0	1.0000
0.2500	0.8828
0.5000	0.8243
0.7500	0.8175
1.0000	0.8565
1.2500	0.9358
1.5000	1.0508
1.7500	1.1971
2.0000	1.3713
2.2500	1.5700
2.5000	1.7903
2.7500	2.0297
3.0000	2.2860

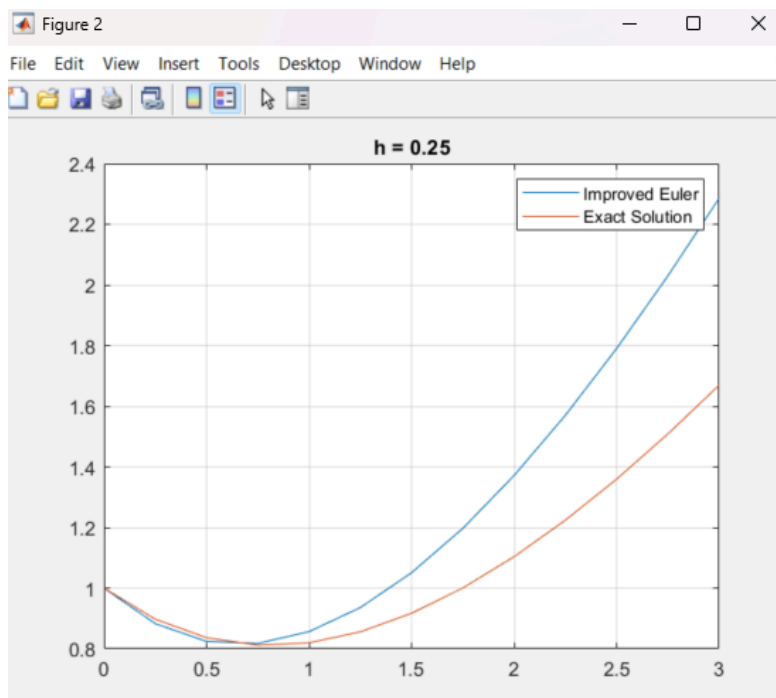
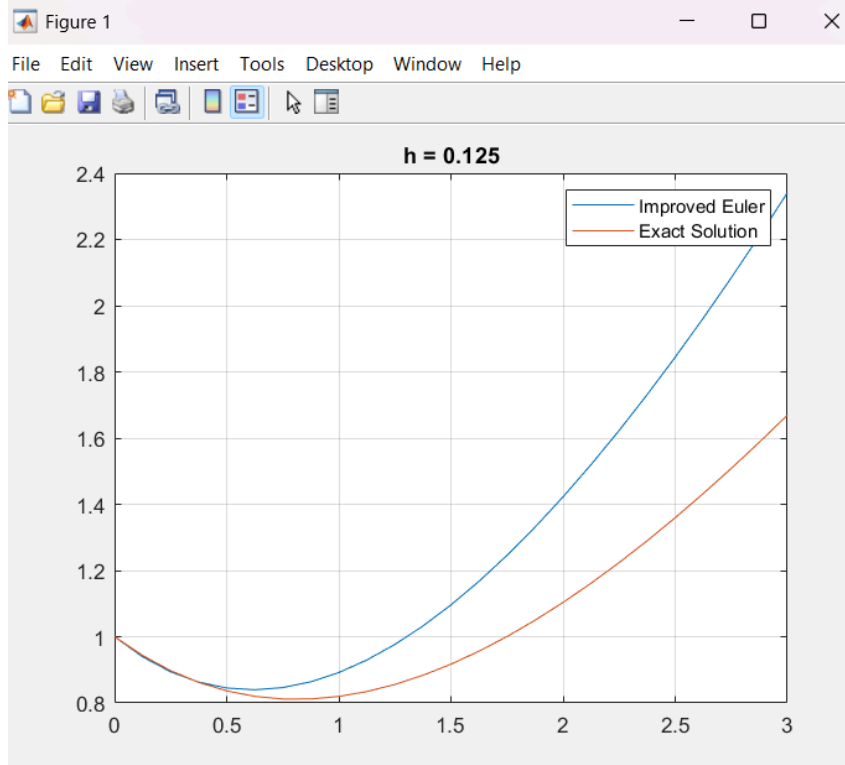
for h = 0.500000

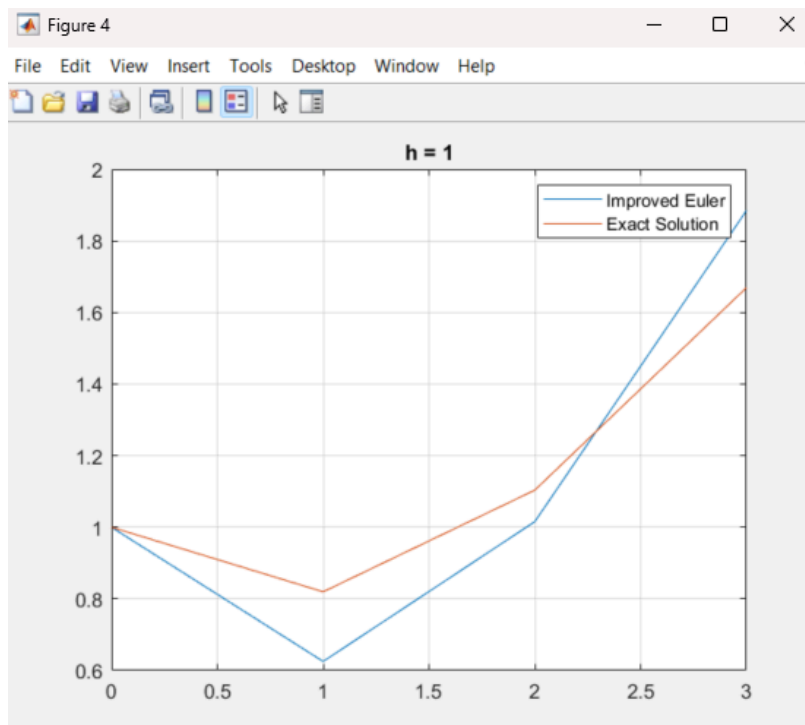
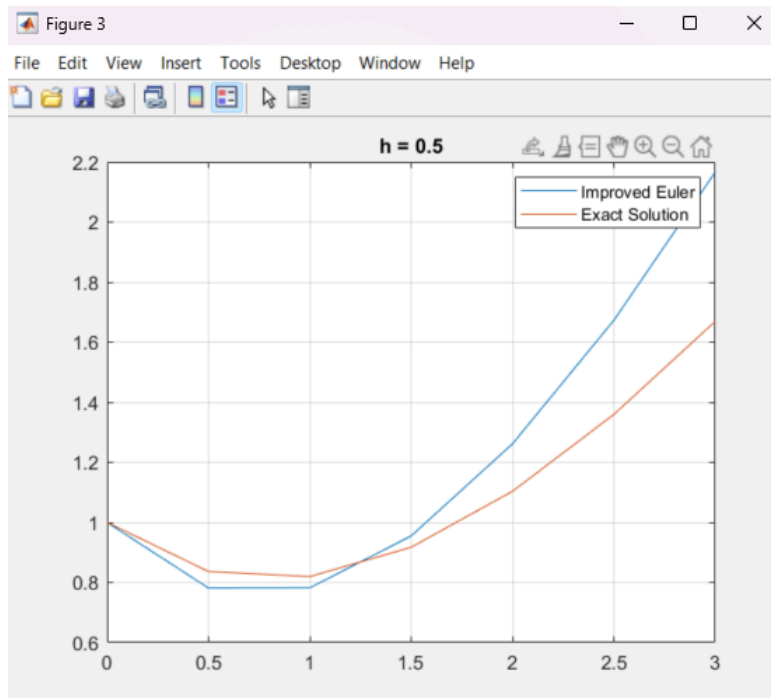
x	Euler y
0	1.0000
0.5000	0.7813
1.0000	0.7822
1.5000	0.9549
2.0000	1.2616
2.5000	1.6731
3.0000	2.1665

for h = 1.000000

x	Euler y
0	1.0000
1.0000	0.6250
2.0000	1.0156
3.0000	1.8848

Graphs: (Graph between Improved Euler and Exact Solution)





Explanation:

This code solves a first-order differential equation using the Improved Euler method and compares it with the exact solution. The program runs the calculation for four different step sizes ($h = 0.125, 0.25, 0.5, 1$) to study accuracy. For each step size, the solution starts from the initial value $y(1) = 1$. First, a predicted value (y_{pred}) is found using the basic Euler formula and then this value is corrected by averaging the slope at the beginning and the predicted point. This makes the method more accurate than Euler's simple method. After computing the numerical values, the code prints the results, calculates the exact solution $y = 3e^{-x/2} - 2 + x$ and plots both solutions. The graphs show that smaller step sizes give results very close to the exact solution, demonstrating the higher accuracy of the Improved Euler method.

Exercise 2(Improved Euler's method)

Code:

```
clear all;
```

```
close all;
```

```
clc;
```

```
R = 20e3;
```

```
C = 10e-6;
```

```
E = 117;
```

```
h = 0.1;          % step size
```

```
t = 0:h:3;
```

```
Q = zeros(1, length(t));
```

```
Q(1) = 0;
```

```
% Euler's Method
```

```
for n = 2:length(t)
```

```
    Q_pred= Q(n-1) + h * (E/R - Q(n-1) / (R*C));
```

```
    Q(n) = Q(n-1) + (h/2)*((E/R - Q(n-1)/(R*C)) + (E/R - Q_pred/(R*C)));
```

```
end
```

```
disp(' t      Euler Q');
```

```
disp([t' Q']);
```

```
% Exact solution
```

```
Q_exact = C*E*(1 - exp(-t/(R*C)));
```

```
% Plot
```

```
plot(t, Q)
```

```
hold on
```

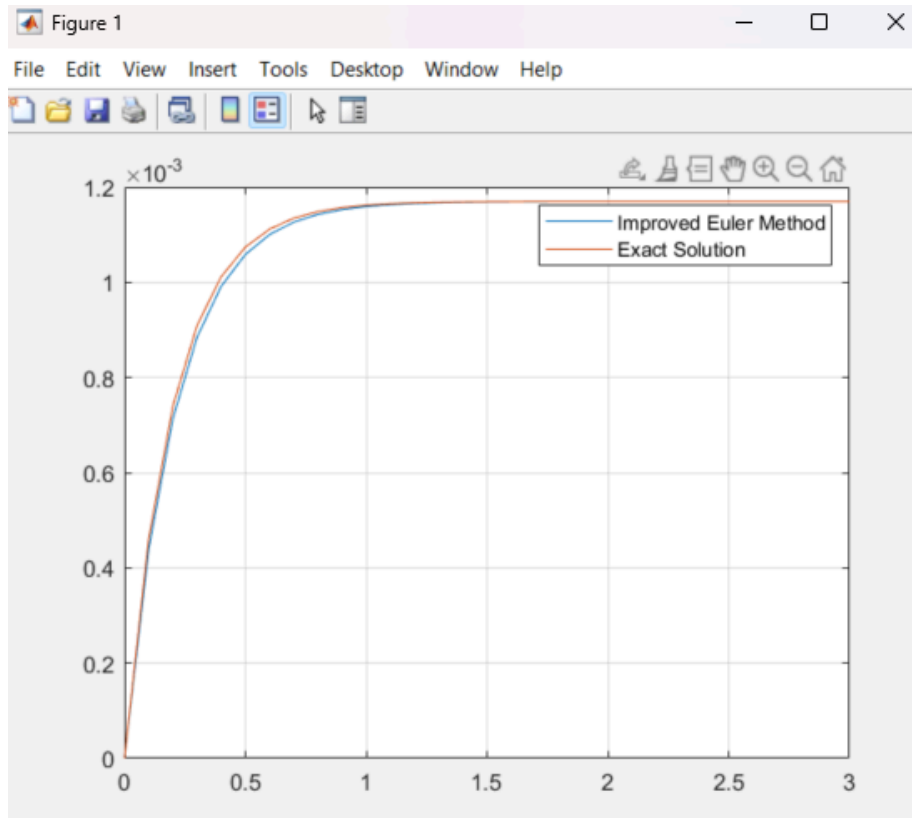
```
plot(t, Q_exact)
```

```
grid on
```

```
legend('Improved Euler Method', 'Exact Solution')
```

Output:

t	Improved Euler Q
0	0
0.1000	0.0004
0.2000	0.0007
0.3000	0.0009
0.4000	0.0010
0.5000	0.0011
0.6000	0.0011
0.7000	0.0011
0.8000	0.0011
0.9000	0.0012
1.0000	0.0012
1.1000	0.0012
1.2000	0.0012
1.3000	0.0012
1.4000	0.0012
1.5000	0.0012
1.6000	0.0012
1.7000	0.0012
1.8000	0.0012
1.9000	0.0012
2.0000	0.0012
2.1000	0.0012
2.2000	0.0012
2.3000	0.0012
2.4000	0.0012
2.5000	0.0012
2.6000	0.0012
2.7000	0.0012
2.8000	0.0012
2.9000	0.0012
3.0000	0.0012



Explanation:

This code uses the Improved Euler method to find the charge $Q(t)$ in an RC circuit and then compares it with the exact solution. First, the resistance R , capacitance C , and voltage E are defined, and a step size $h=0.1$ is chosen. The time range is set from 0 to 3 seconds and the initial charge on the capacitor is zero. At each time step, the code first calculates a predicted charge using the basic Euler formula, and then corrects it by averaging the slope at the current point and the predicted point. This correction improves accuracy compared to the simple Euler method. Finally, the numerical results are displayed and plotted together with the exact solution $Q=CE(1-e^{-t/(RC)})$. The plot shows that the Improved Euler method follows the exact charging curve more closely.

Discussion:

From the results and graphs, it is observed that both Euler's method and the Improved Euler method can approximate the solution of differential equations, but their accuracy depends on the step size. Smaller step sizes produce results closer to the exact solution, while larger step sizes increase error. The Improved Euler method gives better accuracy than the simple Euler method because it uses a correction step. In the RC circuit problem, the numerical solutions correctly show the gradual charging behavior of the capacitor. Overall, this experiment demonstrates that numerical methods are useful for solving practical engineering problems, and choosing a proper step size and method is important for obtaining accurate results.